

Experimental Methods in Computer Science

Departamento de Engenharia Informática, FCTUC, 2025/2026

Experimental Methods in Computer Science (Metodologias Experimentais em Informática)

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Hypothesis Testing *ANalysis Of Variance - ANOVA*

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Problem: what if you have more than two configurations to compare?

Assume you are the database administrator of a big information system. The database has just been installed and you are trying to tune three configurations: Conf. A, Conf. B and Conf. C.

You use a given SQL package to test the execution time for each configuration.

After running several times the SQL package in each configuration, you want to take a decision.

Questions:

- Are the configurations the same?
- What is the best configuration?

Conf. A	Conf. B	Conf. C
exec. time	exec. time	exec. time
74	69	65
66	71	68
88	80	64
68	88	70
79	64	72
68	65	64
87	74	62
79	76	75
78	89	71
72	68	69
86	67	
85	72	
86		

You cannot use a two-sample T-test (at least directly)

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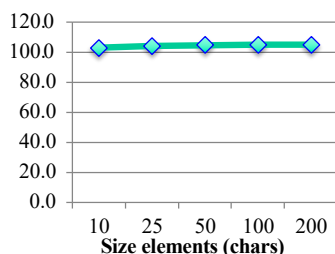
Another example: what is the impact of the size of the elements in the sorting time?

How does the size of the elements in an array (each element is a string of chars) affect the time necessary to sort all the elements of the array?

After performing some experiments with Quicksort and an array of 10000 elements, you obtained the following results:

Does the size of elements matter?

Sorting time



Size of elements (no. characters)				
10	25	50	100	200
106	106	105	108	104
102	105	101	105	105
99	104	106	108	104
103	102	105	104	103
108	101	106	105	108
102	107	107	107	110
104	105	101	104	104
100	106	107	100	102
	102	105	106	105
	103		102	
Means (milliseconds)				
103.0	104.1	104.8	104.9	105.0

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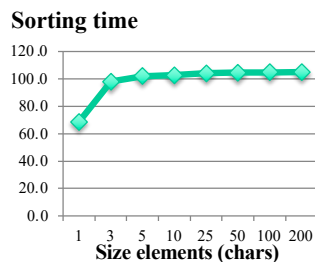
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Another example: what is the impact of the size of the elements in the sorting time?

How does the size of the elements in an array (each element is a string of chars) affect the time necessary to sort all the elements of the array?

After observing the results, you decided to do **some more experiments**:

Does the size of elements matter?



Size of elements (no. characters)							
1	3	5	10	25	50	100	200
71	100	104	106	106	105	108	104
71	95	99	102	105	101	105	105
69	94	103	99	104	106	108	104
69	99	100	103	102	105	104	103
68	99	103	108	101	106	105	108
66	97	100	102	107	107	107	110
71	103	99	104	105	101	104	104
71	98	105	100	106	107	100	102
66	100	103		102	105	106	105
65		105		103		102	

Means (milliseconds)							
68.7	98.3	102.1	103.0	104.1	104.8	104.9	105.0

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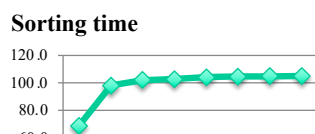
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Another example: what is the impact of the size of the elements in the sorting time?

How does the size of the elements in an array (each element is a string of chars) affect the time necessary to sort all the elements of the array?

After observing the results, you decided to do **some more experiments**:

Does the size of elements matter?



Size of elements (no. characters)							
1	3	5	10	25	50	100	200
71	100	104	106	106	105	108	104
71	95	99	102	105	101	105	105
69	94	103	99	104	106	108	104
69	99	100	103	102	105	104	103
68	99	103	108	101	106	105	108
66	97	100	102	107	107	107	110

But the problem needs more than just looking at the means using an informal approach...

The goal is to test if the difference between multiple sample means is significant.

You need ANOVA to do that!

68.7	98.3	102.1	103.0	104.1	104.8	104.9	105.0
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Informal approach

To know if the difference between multiple sample means is significant...

- Use informal graphic approaches (exploratory data analysis)
 - Charts
 - Side-by-side box plots
 - Multiple histograms
- But knowing whether the differences between the groups (factors) are significant depends on...
 - the difference in the means;
 - the standard deviations of each group;
 - the sample sizes.

**Another distribution
and test like the T or Z**

→ **Need ANOVA to determine P-value (from the F statistic)**

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One-Way ANOVA

- ANalysis Of Variance - **ANOVA**
- The one-way ANOVA is used to test the claim that three or more population means are equal
- This is an extension of the two independent samples tests
- ANOVA tests the following hypotheses:
 - $H_0: \mu_1 = \mu_2 = \mu_3 \dots = \mu_k$ (the means of all the groups are equal)
 - H_1 : Not all the means are equal
- It does not say how or which ones differ. Need to follow up with multiple comparisons (more tests).

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One-Way ANOVA

- **Dependent variable:** the variable you are comparing
- **Independent variable:** the factor variable being used to define the samples (groups)
- **Levels:** values of the independent variable selected to be studied. Each level will originate a sample (group)

- **Example:**
 - Dependent variable: **sorting time**
 - Independent variable: **elements size**
 - Levels: 5 levels
 - 10 chars • 25 chars • 50 chars
 - 100 chars • 200 chars

Size of elements (no. characters)				
10	25	50	100	200
103	108	108	110	108
108	107	109	108	112
105	106	111	111	108
109	104	108	112	113
108	103	109	108	112
103	109	110	110	114
108	107	107	107	108

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Assumptions of ANOVA

- Each group is approximately normal
 - Can be checked informally by looking at the histogram of the data or using a normality test
- Can cope with some non-normality, but not with severe outliers
- Standard deviations of each group are approximately equal
 - Rule of thumb: ratio of largest to smallest sample standard deviation must be less than 2:1

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Rationale for ANOVA

- We have at least 3 means to test (each mean is from a sample). For example, $H_0: \mu_1 = \mu_2 = \mu_3$.
- We could use **two sample t-test** to test them 2 at a time. But with ANOVA we will test them all 3 (or more) at once.
- Instead of using a mean difference, **ANOVA use the variance of the group means in relation to the grand mean over all groups.** Type I error (false positive) also increase dramatically.
- The logic is just the same as for the t-test or z-test: compare the observed variance among means (observed difference in means in the t-test or z-test) to what we would expect to get by chance.

Number of samples	Number of tests
2	1
3	3
4	6
5	10

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- Instead of using a mean difference, **ANOVA use the variance of the group means in relation to the grand mean over all groups.**
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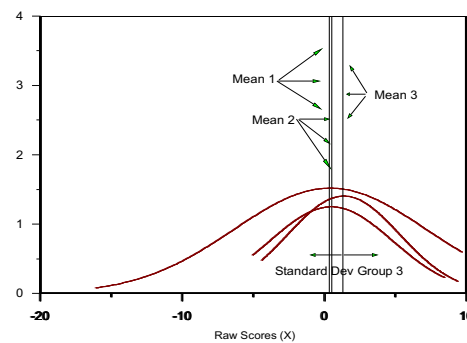
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Rationale for ANOVA (cont.)

Suppose we have 3 samples from the same population. Our results might look like this:

Note that the means from the 3 groups are not exactly the same, but they are close, so **the variance among means will be small.**

Three Samples from the Same Population



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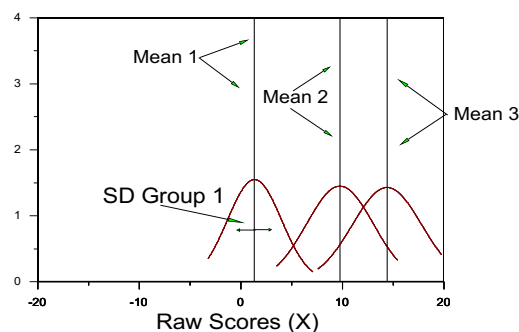
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Rationale for ANOVA (cont.)

Suppose we sample from 3 different populations. Our results might look like this:

Note that the sample means are far away from one another, so **the variance among means will be large.**

Three Samples from 3 Different Populations



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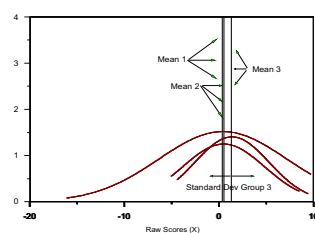
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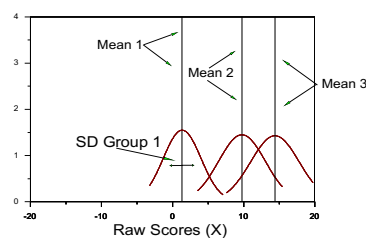
Rationale for ANOVA (cont.)

Suppose we do a study and find the following results (either graph). How would we know or decide whether there is a real effect or not?

Three Samples from the Same Population



Three Samples from 3 Different Populations



To decide, we can compare our observed variance in means to what we would expect to get on the basis of chance, if H_0 is true (i.e., no difference in the means).

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Definitions of terms in ANOVA

Break the analysis of the variance into meaningful pieces that correspond to independent variable (IV) effects and errors.

$\bar{X}_G$ The Grand Mean, taken over all observations.

$\bar{X}_A$ The mean of any level of the IV (group).

$\bar{X}_{A_1}$ The mean of a specific level (1 in this case).

X_i The observation or raw data for the i^{th} measurement.

Variation is the sum of the squares of the deviations between a value and the mean of the sample (group)

$$SS_{(T)} = \sum (X_i - \bar{X}_G)^2$$

Sum of Squares (SS) is often followed by a variable in parentheses such as $SS_{(B)}$ or $SS_{(W)}$ that indicates which sum of squares is referred to:

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Sources of variations

- Sums of squares measure three sources of variation:
 - Groups (variation **between** group means)
 - Error (variation **within** groups)
 - Total (SST = SSG + SSE)
- Degrees of freedom (n observations, k sample means)
 - $df(\text{total}) = n - 1$
 - $df(\text{within}) = n - k$
 - $df(\text{between}) = k - 1$

$$df(\text{total}) = df(\text{between}) + df(\text{within})$$

Size of elements (no. characters)				
10	25	50	100	200
103	108	108	110	108
108	107	109	108	112
105	106	111	111	108
109	104	108	112	113
108	103	109	108	112
103	109	110	110	114
108	107	107	107	108

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ANOVA calculations

$$GM = \frac{\sum n_i \bar{x}_i}{\sum n_i}$$

The grand mean is the weighted average of the individual sample means

$$SS_{(T)} = \sum (X_i - \bar{X}_G)^2$$

The total sum of squares comes from the distance of all the scores to the grand mean. **This is the big total.**

$$SS_{(W)} = \sum df s^2$$

The within-group sum of squares comes from the distance of the scores to the group (level) means. The degrees of freedom are equal to the sum of the individual df for each sample (i.e., number of samples in each group minus 1). **This indicates error.**

$$SS_{(B)} = \sum n_A (\bar{X}_A - \bar{X}_G)^2$$

The between-groups sum of squares represents the distance of the sample means from the grand mean. **This indicates IV effects.**

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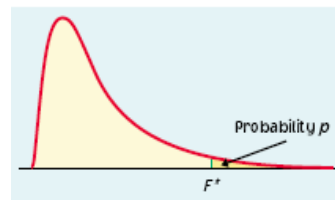
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ANOVA: the F statistics

- The test statistic for ANOVA is called from the **F statistic**, which we obtain from the F Test.
- The **F statistic determines if the variation between sample means is significant**:

Variation Among Sample Means
Variation Among Individuals In Each Sample

$$F = \frac{SS_{(B)} / (k - 1)}{SS_{(w)} / (n - k)}$$



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Obtaining the critical from F table

How to obtain the critical value from F tables for a given α ? For example, for $\alpha = 0.05$?

$$F = \frac{SS_{(B)} / (k - 1)}{SS_{(w)} / (n - k)}$$

df for the numerator

Example: For 4 levels (i.e., 4 values of the IV), $df = k - 1 = 4 - 1 = 3$

df for the denominator

Example: For 13 observations $df = n - k = 13 - 4 = 9$

In the F table for $\alpha = 0.05$, look for the critical value for **$F_{(3,9)}$**

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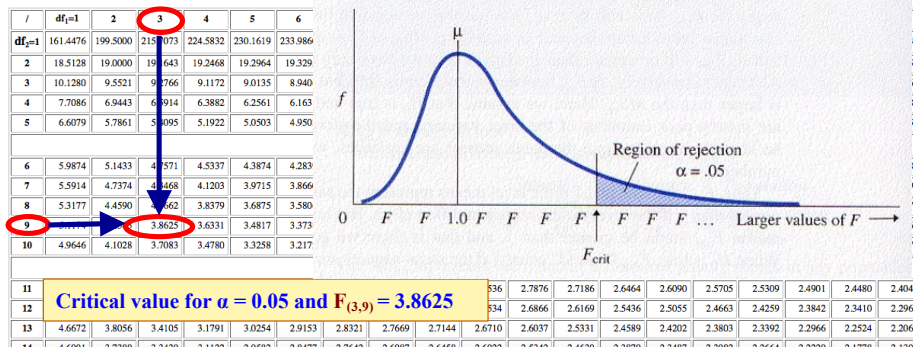
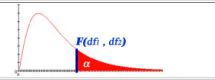
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Obtaining the critical from F table

How

F table for $\alpha = 0.05$



Critical value for $\alpha = 0.05$ and $F_{(3,9)} = 3.8625$

In the F table for $\alpha = 0.05$, look for the critical value for $F_{(3,9)}$

Online F critical value calculator: <https://www.danielsoper.com/statcalc/calculator.aspx?id=4>

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The ANOVA table

- The ANOVA table is a summary of all the elements needed for the calculation of the P-value

	SS	df	MS	F	P
Between	$SS(B)$	$k-1$	$\frac{SS(B)}{k-1}$	$\frac{MS(B)}{MS(W)}$	Tail area above F
Within	$SS(W)$	$n-k$	$\frac{SS(W)}{n-k}$		
Total	$SS(W) + SS(B)$	$n-1$			

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The ANOVA table

- The ANOVA table is a summary of all the elements needed for the calculation of the P-value

	SS	df	MS		P
Between	$SS(B)$	$k-1$	$\frac{SS(B)}{k-1}$	$\frac{MS(B)}{MS(W)}$	Tail area above F
Within	$SS(W)$	$n-k$	$\frac{SS(W)}{n-k}$		
Total	$SS(W) + SS(B)$				

$$SS_{(B)} = \sum n_A (\bar{X}_A - \bar{X}_G)^2$$

$$SS_{(W)} = \sum df s^2$$

$$GM = \frac{\sum n_i \bar{x}_i}{\sum n_i}$$

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One-way ANOVA steps

Use the same general steps of hypothesis testing

1. State the hypothesis
2. Calculations (to compute the test statistic)
3. Obtain p value
4. Make a decision

$$F = \frac{SS_{(B)} / (k - 1)}{SS_{(W)} / (n - k)}$$

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Example one-way ANOVA

How does the size of the elements in an array affect the time necessary to sort all the elements of the array?

For economy of space, we will consider only 3 level of the independent variable size of elements: 3, 5 and 10 characters.

The table on the right shows the sorting times in milliseconds obtained with the Quicksort and an array of 10000 elements.

Size of elements (no. characters)		
3	5	10
100	104	106
95	99	102
94	103	99
99	100	103
99	103	108
97	100	102
103	99	104
98	105	100
100	103	
	105	

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Example: one-way ANOVA

Step 1- State the hypothesis

Hypothesis

- **H0: $\mu_3 = \mu_5 = \mu_{10}$** → All samples have equal means
- **H1: $\mu_3 \neq \mu_5 \neq \mu_{10}$** → Not all the means are equal
 - Does not say how or which ones differ
 - Can follow up with multiple comparisons

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Example: one-way ANOVA

Step 2 - Calculations

Determine characteristics of the samples under comparison.

	3 chars	5 chars	10 chars
Sample size	9	10	8
Mean	98.3	102.1	103.0
Std. Dev.	2.74	2.38	2.98
Variance	7.50	5.66	8.86

$$n = 27$$

$$k = 3$$

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Example: one-way ANOVA

Step 2 – Calculations (cont.)

Grand Mean calculation

- Weighted average of the individual sample means

$$GM = \frac{\sum n_i \bar{x}_i}{\sum n_i}$$

	3 chars	5 chars	10 chars
Sample size	9	10	8
Mean	98.3	102.1	103.0
Std. Dev.	2.74	2.38	2.98
Variance	7.50	5.66	8.86

$$GM = \frac{9(98.3) + 10(102.1) + 8(103.0)}{9 + 10 + 8} = 101.10$$

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Example: one-way ANOVA

Step 2 – Calculations (cont.)

Between Group Variation

- Variation between each sample mean and the grand mean
- Each group variation is weighted by the sample size

$$SS_{(B)} = \sum n_A (\bar{X}_A - \bar{X}_G)^2$$

$$SS(B) = 9(98.3 - 101.10)^2 + 10(102.1 - 101.10)^2 + 8(103.0 - 101.10)^2 = 107.77$$

$$GM = 101.10$$

	3 chars	5 chars	10 chars
Sample size	9	10	8
Mean	98.3	102.1	103.0
Std. Dev.	2.74	2.38	2.98
Variance	7.50	5.66	8.86

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Example: one-way ANOVA

Step 2 – Calculations (cont.)

Within Group Variation

- Is the weighted total of the individual variations
- The weighting is done with the degrees of freedom. For each sample df is one less than the sample size

$$SS_{(W)} = \sum df s^2$$

	3 chars	5 chars	10 chars
Sample size	9	10	8
Mean	98.3	102.1	103.0
Std. Dev.	2.74	2.38	2.98
Variance	7.50	5.66	8.86

$$SS(W) = 8(2.74)^2 + 9(2.38)^2 + 7(2.98)^2 = 172.90$$

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Example: one-way ANOVA Step 2 – Calculations (cont.)

We can now fill the one-way ANOVA table

	SS	df	MS	F	P
Between	SS(B)	k-1	$\frac{SS(B)}{k-1}$	$\frac{MS(B)}{MS(W)}$	Tail area above F
Within	SS(W)	n-k	$\frac{SS(W)}{n-k}$		
Total	SS(W) + SS(B)	n-1			

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Example: one-way ANOVA Step 2 – Calculations (cont.)

We can now fill the one-way ANOVA table

	SS	df	MS	F	P
Between	107.77	2	53.88	7.48	≈ 0.003
Within	172.90	24	7.20		
Total	280.67	26			

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Example: one-way ANOVA

Step 3 – Obtain P-value

We can now fill the one-way ANOVA table

	SS	df	MS	F	P
Between	107.77	2	53.88	7.48	≈ 0.003
Within	172.90	24	7.20		
Total	280.67	26			

The P-Value for $F_{(2,24)} = 7.48$ is 0.002986 (using an online calculator)

<https://www.socscistatistics.com/pvalues/fdistribution/calculator/>

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Example: one-way ANOVA

Step 4 – Make a decision

	SS	df	MS	F	P
Between	107.77	2	53.88	7.48	≈ 0.003
Within	172.90	24	7.20		
Total	280.67	26			

$H_0: \mu_3 = \mu_5 = \mu_{10}$
means

→ All samples have equal

$P \approx 0.003 \rightarrow H_0$ is Rejected!

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Using an online one-way ANOVA calculator

Analysis of Variance (ANOVA) Calculator - One-Way ANOVA from Summary Data

This calculator will generate a complete one-way analysis of variance (ANOVA) table for up to 10 groups, including sums of squares, degrees of freedom, mean squares, and F and p-values, given the mean, standard deviation, and number of subjects in each group.

Please enter the necessary parameter values, and then click 'Calculate'.

<https://www.danielsoper.com/statcalc/calculator.aspx?id=43>

	Number of Subjects	Mean	Standard Deviation
Group 1:	9	98.3	2.74
Group 2:	10	102.1	2.38
Group 3:	8	103.0	2.98
Group 4:			
Group 5:			
Group 6:			
Group 7:			
Group 8:			
Group 9:			
Group 10:			

Calculate!

	SS	df	MS	F	p
Between:	109.440	2	54.720	7.582	0.003
Within:	173.203	24	7.217		
Total:	282.643	26			

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Two-Way ANOVA

- Tests the equality of two or more population means when two independent variables are used: **factor A** and **factor B** (more than two factors: multi-way ANOVA).
- Each independent variables (factors) may have any number of levels.
- Same results as separate one-way ANOVA on each variable. **But interaction can be tested.**
- Saves time and effort, compared to consecutive one-way ANOVA tests.

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Two-way ANOVA assumptions

- Normality
Populations are normally distributed
- Homogeneity of variance
Populations have similar variances
- Independence of errors
Independent random samples

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Two-way ANOVA: Null Hypotheses

Tests 3 hypothesis simultaneously:

- No difference in means due to factor A
– $H_0: \mu_{1.} = \mu_{2.} = \dots = \mu_{a.}$
- No difference in means due to factor B
– $H_0: \mu_{.1} = \mu_{.2} = \dots = \mu_{.b}$
- No interaction of factors A and B
– $H_0: AB_{ij} = 0$

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Two-way ANOVA vs. one-way ANOVA

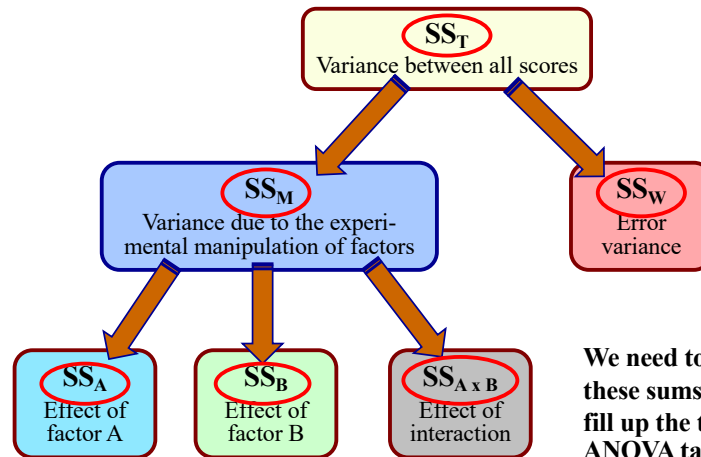
- The test follows the same steps in both cases.

1. State the hypothesis
2. Calculations (to compute the test statistic)
3. Obtain p value
4. Make a decision

Two-way ANOVA vs. one-way ANOVA

- The test follows the same steps in both cases.
- The two-way ANOVA table is similar (but has more lines than the one-way ANOVA table as there are more sums of squares and more degrees of freedom to calculate).
- The formulas for the calculations in two-way ANOVA are similar to the ones used in one-way ANOVA.
- The F metric is the same.
- The interpretation of the results (two-way ANOVA table filled up with the calculation is more complex, but still quite straightforward).

Two-way ANOVA sums of squares (SS)



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Two-way ANOVA table

	SS	df	MS	F	Decision
Factor A	$SS_{(A)}$	$a-1$	$\frac{SS_{(A)}}{a-1}$	$\frac{MS_{(A)}}{MS_{(W)}}$	Tail area above F
Factor B	$SS_{(B)}$	$b-1$	$\frac{SS_{(B)}}{b-1}$	$\frac{MS_{(B)}}{MS_{(W)}}$	Tail area above F
Interaction	$SS_{(A \times B)}$	$(a-1)(b-1)$	$\frac{SS_{(A \times B)}}{(a-1)(b-1)}$	$\frac{MS_{(A \times B)}}{MS_{(W)}}$	Tail area above F
Within (error)	$SS_{(W)}$	$n-a-b$	$\frac{SS_{(W)}}{n-a-b}$		
Total	$SS_{(M)} + SS_{(W)}$	$n-1$			

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Two-way ANOVA table

	SS	df	MS	F	Decision
Factor A	$SS_{(A)}$	.	$SS_{(A)}$	$MS_{(A)}$	Tail area
Factor B	$SS_{(B)}$				
Interaction	$SS_{(A \times B)}$				
Within (error)	$SS_{(W)}$				
Total	$SS_{(M)} + SS_{(W)}$				

Diagram illustrating the decomposition of total variance (SS_T) into between-group variance (SS_M) and within-group variance (SS_W). SS_M is further decomposed into the effects of factor A (SS_A), factor B (SS_B), and their interaction ($SS_{A \times B}$).

We need to calculate all these sums of squares to fill up the two-way ANOVA table

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Example two-way ANOVA

How does the **number of elements in an array** and **the size of each element** affect the sorting time of the array?

For economy of space, we will consider only 3 level of the independent variable **size of element** (10, 50 and 100 characters) and 2 levels of the independent variable **number of elements** (10000 and 50000).

The table on the right shows the sorting times in milliseconds obtained with a Quicksort implementation.

10 chars		50 chars		100 chars	
10K	50K	10K	50K	10K	50K
106	313	105	308	108	312
102	307	101	309	105	307
103	308	106	307	108	307
103	309	105	307	104	311
108	306	106	311	105	312
102	308	107	310	107	308
104	310	101	311	104	308
105	312	107	311	100	309
108		105		106	312
				102	

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Example: two-way ANOVA

Step 1- State the hypothesis

Factor A – Size (in chars) of the elements to be sorted

Factor B – Number of elements to sort

Hypothesis

- $H_0: \mu_{a,10} = \mu_{a,50} = \mu_{a,100} \rightarrow$ The size of elements to be sorted is not relevant for the sorting time (means are equal)
- $H_0: \mu_{b,10k} = \mu_{b,100K} \rightarrow$ The number of elements to be sorted is not relevant for the sorting time (means are equal)
- $H_0: AB_{ij} = 0 \rightarrow$ There are no interaction between the size and the number of elements
- $H_1: \rightarrow$ Not all the means are equal and there is interaction (only right tail is possible).

Example: two-way ANOVA

Step 2 - Calculations

Determine characteristics of the samples under comparison.

	10 chars		50 chars		100 chars	
	10K	50K	10K	50K	10K	50K
Sample size	9	8	9	8	10	9
Mean	104.1	308.8	104.8	309.3	104.9	309.6
Std. Dev.	1.96	2.12	2.28	1.75	2.56	2.19
Variance	3.86	4.50	5.19	3.07	6.54	4.78

$$GM = \frac{\sum n_i \bar{x}_i}{\sum n_i} \quad \text{Weighted average of the individual sample means}$$

$$GM = \frac{9(104.1) + 8(308.8) + 9(104.8) + 8(309.3) + 10(104.9) + 9(309.6)}{9 + 8 + 9 + 8 + 10 + 9} = 201.25$$

Example: two-way ANOVA

Step 2 – Calculations (cont.)

$SS_{(T)}$ - The total sum of squares comes from the distance of all the scores from the grand mean. This is the big total

$$SS_{(T)} = \sum (X_i - \bar{X}_G)^2$$

$\bar{X}_G$ is the GM (Grand Mean), taken over all observations/scores.

The observation or raw data for the i^{th} measurement/score

Performing the calculation: $SS_{(T)} = 552982$

(actually, this value is not needed in the calculations)

Example: two-way ANOVA

Step 2 – Calculations (cont.)

$SS_{(M)}$ – The sum of squares that go to the experimental manipulation of factor

$$SS_{(M)} = \sum n_i (\bar{X}_i - \bar{X}_G)^2$$

Number of samples of level i

Average of the scores of level i

	10 chars		50 chars		100 chars	
	10K	50K	10K	50K	10K	50K
Sample size	9	8	9	8	10	9
Mean	104.1	308.8	104.8	309.3	104.9	309.6
Std. Dev.	1.96	2.12	2.28	1.75	2.56	2.19
Variance	3.86	4.50	5.19	3.07	6.54	4.78

Performing the calculation:

$$SS_{(M)} = 9(104.1 - 201.25)^2 + 8(308.8 - 201.25)^2 + 9(104.8 - 201.25)^2 + 8(309.3 - 201.25)^2 + 10(104.9 - 201.25)^2 + 9(309.6 - 201.25)^2 = 552736.78$$

Example: two-way ANOVA

Step 2 – Calculations (cont.)

$SS_{(A)}$ – The sum of squares for factor A (effect of factor A)

$$SS_{(A)} = \sum n_i (\bar{X}_i - \bar{X}_G)^2$$

First, organize the data according factor A observations

10 chars	
10K	50K
106	313
102	307
103	308
103	309
108	306
102	308
104	310
105	312
108	

50 chars	
10K	50K
105	308
101	309
106	307
105	307
106	311
107	310
101	311
107	311
105	

100 chars	
10K	50K
108	312
105	307
108	307
104	311
105	312
107	308
104	308
100	309
106	312
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Example: two-way ANOVA

Step 2 – Calculations (cont.)

$SS_{(A)}$ – The sum of squares for factor A (effect of factor A, size of each element to be sorted)

$$SS_{(A)} = \sum n_i (\bar{X}_i - \bar{X}_G)^2$$

$$SS_{(A)} = 17(200.82 - 201.25)^2 + 17(201.00 - 201.25)^2 + 19(201.84 - 201.25)^2 = 10.81$$

Example: two-way ANOVA

Step 2 – Calculations (cont.)

$SS_{(B)}$ – The sum of squares for factor B (effect of factor B)

$$SS_{(B)} = \sum n_i (\bar{X}_i - \bar{X}_G)^2$$

Organize the data according
factor B observations

10K		
10 chars	50 chars	100 chars
106	105	108
102	101	105
103	106	108
103	105	104
108	106	105
102	107	107
104	101	104
105	107	100
108	105	106
		102

50K		
10 chars	50 chars	100 chars
313	308	312
307	309	307
308	307	307
309	307	311
306	311	312
308	310	308
310	311	308
312	311	309
		312

Compute
averages

Example: two-way ANOVA

Step 2 – Calculations (cont.)

$SS_{(B)}$ – The sum of squares for factor B (effect of factor B,
number of elements to be sorted)

$$SS_{(B)} = \sum n_i (\bar{X}_i - \bar{X}_G)^2$$

$$SS_{(B)} = 28(104.75 - 201.25)^2 + 25(309.32 - 201.25)^2 = 552721.12$$

Example: two-way ANOVA

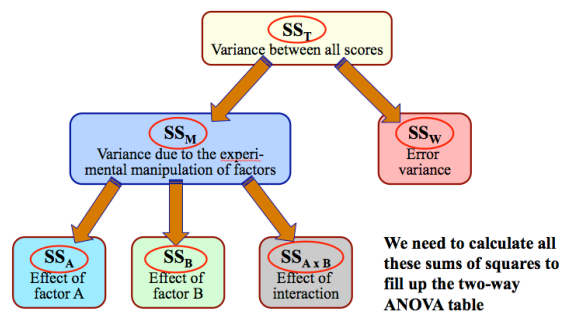
Step 2 – Calculations (cont.)

$SS_{(A \times B)}$ – The sum of squares for the interaction between factor A and factor B

$$SS_{(A \times B)} = SS_M - SS_A - SS_B$$

Performing the calculation:

$$SS_{(A \times B)} = 552736.78 - 10.81$$



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Example: two-way ANOVA

Step 2 – Calculations (cont.)

Within Group Variation

- Is the weighted total of the inc
- The weighting is done with the degrees of freedom. For each sample df is one less than the sample size

	10 chars		50 chars		100 chars	
	10K	50K	10K	50K	10K	50K
Sample size	9	8	9	8	10	9
Mean	104.1	308.8	104.8	309.3	104.9	309.6
Std. Dev.	1.96	2.12	2.28	1.75	2.56	2.19
Variance	3.86	4.50	5.19	3.07	6.54	4.78

$$SS_{(W)} = \sum df s^2$$

$$SS_{(W)} = 1.96^2 (9 - 1) + 2.12^2 (8 - 1) + 2.28^2 (9 - 1) + 1.75^2 (8 - 1) + 2.56^2 (10 - 1) + 2.19^2 (9 - 1) = 245.28$$

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Example: two-way ANOVA

Step 2 – Calculations (cont.)

Summary table:

	SS	df	MS	F	Decision
Factor A	$SS_{(A)}$	$a-1$	$\frac{SS_{(A)}}{a-1}$	$\frac{MS_{(A)}}{MS_{(W)}}$	Tail area above F
Factor B	$SS_{(B)}$	$b-1$	$\frac{SS_{(B)}}{b-1}$	$\frac{MS_{(B)}}{MS_{(W)}}$	Tail area above F
Interaction	$SS_{(A \times B)}$	$(a-1)(b-1)$	$\frac{SS_{(A \times B)}}{(a-1)(b-1)}$	$\frac{MS_{(A \times B)}}{MS_{(W)}}$	Tail area above F
Within (error)	$SS_{(W)}$	$n-a-b$	$\frac{SS_{(W)}}{n-a-b}$		
Total	$SS_{(M)} + SS_{(W)}$	$n-1$			

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Example: two-way ANOVA

Step 2 – Calculations (cont.)

Summary table:

	SS	df	MS	F	Decision
Factor A	10.81	2	5.41	1.06	Tail area above F
Factor B	552721.12	1	552721.12	108166.86	Tail area above F
Interaction	4.84	2	2.42	0.47	Tail area above F
Within (error)	245.28	48	5.11		
Total	552982.05	52			

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Example: two-way ANOVA

Step 4 – Make a decision

Summary table:

	SS	df	MS	F	Decision
Factor	Hypothesis • $H_0: \mu_{b,10} = \mu_{b,50} = \mu_{b,100} \rightarrow$ The size of elements to be sorted is not relevant for the sorting time (means are equal) • $H_0: \mu_{10k,a} = \mu_{50K,a} \rightarrow$ The number of elements to be sorted is not relevant for the sorting time (means are equal) • $H_0: AB_{ij} = 0 \rightarrow$ There are no interaction between the size and the number of elements • $H_1: \rightarrow$ Not all the means are equal and there is interaction (only right tail is possible).				
Factor					
Interac					
Within					
Total					

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Example: two-way ANOVA

Step 2 – Calculations (cont.)

Summary table:

	SS	df	MS	F	Decision
Factor A	10.81	2	5.41	1.06	0.3589
Factor B	44.96	2	22.48	4.49	≈ 0
Interaction	4.84	2	2.42	0.47	0.6252
Within (error)	24	24	1		
Total	5529.62105	32			

P value = 36% for the factor A, size of elements to be sorted $\rightarrow H_0$ is retained for this factor: the size of the elements is not relevant for the sorting time

P value ≈ 0 for the factor B, number of elements to be sorted $\rightarrow H_0$ is rejected: the number of elements to be sorted is significant and determines the sorting time.

P value = 63% for the interactions between factor A and B $\rightarrow H_0$ is retained: there is no interactions between the size of the elements and the number of elements

H_0 as whole is rejected

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Examples of online calculators and etc.

- **Example of F critical value calculator:**
<http://www.danielsoper.com/statcalc3/calc.aspx?id=4>
- **F critical tables:**
http://www.socr.ucla.edu/applets.dir/f_table.html
- **ANOVA calculator:**
<http://www.danielsoper.com/statcalc3/calc.aspx?id=43>
- **Two-way ANOVA calculator:**
<http://scistatcalc.blogspot.pt/2013/11/two-factor-anova-test-calculator.html#>
- **P calculator for several distributions:**
<http://vassarstats.net/tabs.html>
- **Videos two-way ANOVA**
<https://www.youtube.com/watch?v=cNIIIn9bConY&list=PLWtoq-EhUJe2TjJYfZUQtuq7a0dQCnOWp>